

Aufgabe 3

The screenshot displays the IntelliJ IDEA IDE with the following components:

- Project View (Left):** Shows the project structure for 'team92'. The 'src' directory contains 'LibraryCLI.java' and 'LibraryManagementTests.java'. The 'LibraryCLI.java' file is highlighted, showing its coverage status as 0%.
- Code Editor (Center):** Displays the source code of 'LibraryCLI.java'. The code includes imports for 'java.util.*', 'BookManager', 'BookCopyManager', 'CustomerManager', 'LoanService', and 'Scanner'. It defines a 'LibraryCLI' class with private final fields for these managers and services, a constructor, a 'main' method, and a 'start' method.
- Coverage Table (Right):** Titled 'Coverage BookManagerTest and 2 more', it provides a detailed breakdown of coverage metrics for various elements in the project.

Element	Class, %	Method, %	Line, %	Branch, %
all	81% (9/11)	50% (15/30)	57% (119/207)	42% (21/50)
Book	100% (1/1)	50% (1/2)	88% (8/9)	100% (0/0)
BookCopy	100% (1/1)	50% (1/2)	87% (7/8)	100% (0/0)
BookCopyManager	100% (1/1)	66% (2/3)	95% (20/21)	87% (7/8)
BookCopyManagerTest	100% (1/1)	100% (2/2)	100% (12/12)	100% (0/0)
BookManager	100% (1/1)	66% (2/3)	94% (18/19)	87% (7/8)
BookManagerTest	100% (1/1)	100% (2/2)	100% (13/13)	100% (0/0)
Customer	100% (1/1)	33% (1/3)	81% (9/11)	100% (0/0)
CustomerManager	100% (1/1)	66% (2/3)	95% (20/21)	87% (7/8)
CustomerManagerTest	100% (1/1)	100% (2/2)	100% (12/12)	100% (0/0)
LibraryCLI	0% (0/1)	0% (0/6)	0% (0/55)	0% (0/16)
LoanService	0% (0/1)	0% (0/2)	0% (0/26)	0% (0/10)

IntelliJ IDEA's integrierte Code-Coverage-Funktionalität für die Testabdeckung.
Alle Unit-Tests laufen unter JUnit 5.

Gesamtübersicht

- Class Coverage: 81 % (9/11 Klassen)
- Method Coverage: 50 % (15/30 Methoden)
- Line Coverage: 57 % (119/2607 Zeilen)
- Branch Coverage: 42 % (21/50 Branches)

Analyse der Top- und Flop-Klassen

Top-Klassen (hohe Abdeckung)

- LibraryManagementTests:

100 % in allen Metriken (Testklasse selbst)

- BookManager, BookCopyManager, LoanService:

Alle wesentlichen Pfade und Fehlersituationen sind abgedeckt.

Flop-Klassen (niedrige Abdeckung)

- LibraryCLI (0 %)

CLI-Logik (Eingabe/Ausgabe über Scanner/System.out) bisher nicht automatisiert getestet.

- Customer (33 % Method Coverage)

Nur der Konstruktor und Getter getFirstName/getId getestet. Weitere Methoden fehlen.

Ursachen für Coverage-Lücken

CLI-Komponenten sind schwieriger zu automatisieren, da sie auf System.in und System.out beruhen.

Einige einfache Getter/Setter wurden bislang nicht einzeln getestet, da sie trivial erscheinen.

Fehlende Branch-Tests in LoanService (z. B. Fälle „-1 zum Abbrechen“, ungültige IDs) können ergänzt werden.